

### 3

## Action, Intention, and the Sense of Agency

Whatever our first action might be, it almost certainly emerges out of our early interactions with others. Moreover, to the extent that we learn to act in specific ways, and that our actions aim at some goal, we learn to act in contexts of interaction, and we learn from others what counts as possible and preferred goals. We don't act or learn to act first, and only then enter into interactions. Rather, social dimensions are built into action from the very start. Accordingly, interaction has a primacy both when it comes to understanding others and in regard to our own actions. I'll present evidence in support of this idea in this chapter.

Philosophical accounts of action often ignore the interactional or social aspects of action. Such accounts are characterized by methodological individualism. That is, the focus tends to be on processes within the individual that lead to or follow from action: intention formation, the sense of agency, motor-control mechanisms, or free will understood as characteristic of the individual person. It is undeniable that some conception of an individual self has to enter into the analysis of action, but I will argue that the self in the "landscape of action" (Bruner 1986) always needs to be understood as located as well in what we might call the landscape of interaction. This is reflected not only in narratives and communicative practices, but also in the actual doings that we call actions.

Both reflective introspection and neuroscientific accounts of the self abstract from or ignore the situated agent as she is normally embedded in contextualized social situations, and this can lead to surprising and seemingly counter-intuitive results. Famously, when David Hume (1739) looks closely and introspectively within his own individual theater of experience, without regard for action or how we engage with others, he finds nothing that could count as a self. And more recently, when Thomas Metzinger (2004) explores the workings of the individual brain, substituting the tools of neuroscience for introspection, he concludes that "no such things as selves exist in the world: Nobody ever *was* or *had* a self" (p. 1). At best we find a "self-model"—a phenomenal experience constituted in neural representations. In this model, action is treated in terms of motor control, and the social is reducible to certain brain processes—notably, simulation processes that, in principle, could be instantiated in a disembodied brain in a vat (2004, 363). Likewise, when Galen Strawson (1997; 2009) sets agency and social relations aside in his account of the self, he ends up equating the self with a momentary abstract existence, seemingly without memory or possibilities.

In the first part of this chapter I focus on the sense of self-agency. I show two things: first, that the phenomenology of the sense of agency is both complex and ambiguous. By looking at some recent studies on the experience of self-agency we can see that this experience is complex. I'll argue that although there surely is some degree of ambiguity in the conceptual analysis of this experience, perhaps because many of the theoretical and empirical studies cut across disciplinary lines, there is also a genuine ambiguity in the experience of self-agency itself. Second, most studies of the sense of agency fail to take into consideration the social dimension. This social dimension accounts for some of the ambiguity.

### 3.1 Initial Complexities in the Sense of Agency

I start with the following question: What is action if it is something more than bodily movement? My answer is that action, which is more than bodily movement and which scales up to more than a set of causal processes, is characterized by three elements or aspects beyond the movement itself: intention, a sense of agency, and meaning which generally goes beyond the agent's intention. Action, in this sense, cannot be unintentional, although there can be unintentional consequences to actions. I'll argue that these aspects are involved in most (although not all) actions, and are, despite their close relation to one another, irreducible to one another, and irreducible to purely causal or mechanical or neural processes.

Ordinarily, in everyday experience, when I engage in action I have a sense of agency for that action. How is that sense or experience of agency generated? It turns out that there are a number of things that can contribute to this experience. In every case, some, but not necessarily all of these things, are integrated into the experience of agency. I'll start with what seems most basic.

#### 3.1.1 *Motor-control processes*

We can start with the idea that action, whether relatively direct and simple (as when I reach and pick up a glass to take a drink), or indirect and complex (as when I author a book, or lead an army to war), whether locomotive, instrumental, or communicative, usually involves intentional bodily movement of some sort. Not always, since, arguably, there are exceptions. If we consider thinking to be an action, I can sit very still and, without moving, multiply four times seven "in my head." I may be able to do this quicker, however, if I'm allowed to gesture with my hands (see Goldin-Meadow et al. 2001). A person with Locked In Syndrome, incapable of bodily movement, may, with the help of a brain-computer interface, control a cursor on a computer screen just by thinking. Usually, however, even thinking, as a form of action, involves movement—sometimes speech and gesture, sometimes assuming a specific bodily posture, sometimes wrinkling one's brow, making facial expressions, or nodding one's head. Such movements may contribute to the accomplishment of thought, as has been shown with gesture (Goldin-Meadow 1999; Gallagher 2005a).

If we think of the sense of agency as the experience that I am the one who is generating and/or controlling my action, then we can distinguish the sense of agency from the sense of ownership. The sense of ownership, or what phenomenologists call

the sense of mineness, applies to the most basic and pre-intentional aspects of action (aspects such as movement and experience), and then scales up to apply immediately to action itself. Accordingly, I can say that this is *my* movement (“I am the one moving”), or this is *my* experience, just as well as I can say that this is *my* action or *my* thinking, and so forth. When my body moves, for example, I have a proprioceptive/kinesthetic experience of that movement, and I can easily sense that it is my body moving; this is the case with either voluntary or involuntary movement (Gallagher 2000a). In the case of voluntary movement or intentional action the sense of agency and the sense of ownership cannot be easily distinguished, and in fact they tend to reinforce each other. In the case of involuntary movement, however, they can be dissociated since for involuntary movement a sense of agency is missing, but the sense of ownership is still involved. If somebody pushes me, for example, I still have the sense that I am the one moving, even if I did not cause the initial movement or have immediate control of it. Sense of agency and sense of ownership also depend on distinguishable neural systems and can be manipulated independently of each other (Tsakiris, Haggard, & Longo 2010).

In terms of phenomenology, sense of agency and sense of ownership, as we are defining them here, are pre-reflective, which means that they neither are equivalent to nor depend on the agent taking an introspective reflective attitude. Nor do they require that the agent engage in an explicit perceptual monitoring of bodily movements. Just as I do not usually attend to the motor details of my own bodily movements as I am engaged in action, my sense of agency, which may involve an experience of control over my movement, is not normally something that I attend to or something of which I am explicitly aware. In this respect sense of agency and sense of ownership, as pre-reflective, are phenomenologically recessive.

Action as self-generated physical movement involves motor-control processes. These include efferent brain processes involved in issuing a motor command. Let’s think again about involuntary movement. In the case of involuntary movement there is a sense of ownership for the initial movement but no sense of agency. I feel it to be *my* movement, in the sense that my body is moving; but I do not feel that I am causing the movement. Awareness of my voluntary or involuntary movement comes from reafferent sensory feedback (visual and proprioceptive/kinesthetic information that tells me that I’m moving). The “proprio” in proprioception is just that self-specific aspect of the process. Thus, in both involuntary and voluntary movement the sense of ownership is generated by these sensory-feedback processes. In the case of involuntary movement, however, there are no initial motor commands (no efferent signals) issued to generate the initial movement. Accordingly, one hypothesis is that in the case of voluntary movement a basic, pre-reflective sense of agency is generated by efferent signals. Tsakiris and Haggard (2005) review empirical evidence to support this division of labor. Sensory-suppression experiments suggest that the sense of agency arises at an early efferent stage in the initiation of action and that awareness of the initiation of my own action depends on pre-motor processes which precede actual bodily movement (Tsakiris & Haggard 2003). Experiments with subjects who lack proprioception but still experience a sense of effort reinforce this conclusion (Lafargue, Paillard, Lamarre, & Sirigu 2003; see Marcel 2003). As Tsakiris and Haggard put it,

[...] the sense of agency involves a strong *effluent* component, because actions are centrally generated. The sense of ownership involves a strong *afferent* component, because the content of body awareness originates mostly by the plurality of multisensory peripheral signals. We do not normally experience the efferent and afferent components separately. Instead, we have a general awareness of our body that involves both components.

(Tsakiris and Haggard 2005, 387)

Indeed, to this last point, they also suggest that efferent processes underlying the sense of agency modulate sensory feedback resulting from movement, supporting the idea that the sense of agency and sense of ownership are closely integrated in voluntary action. The pre-reflective sense of agency arises not simply when I initiate an action; rather, as I continue to control my action, continuing efferent signals, and the afferent feedback that I get from my movement, contribute to an ongoing sense of agency, which may include a sense of effort.

Also note that to the extent that I am aware of my action I tend to be aware of *what* I am doing rather than the details of *how* I am doing it; e.g., what muscles I am using. Even my awareness of what I am doing is struck at the most pragmatic level of description, or at the level of what we called, in *Chapter 1*, the highest realized affordance (e.g., “I’m getting a drink”) rather than at a level of motor-control mechanisms. That is, the motor-control aspects of the sense of agency (and the sense of ownership) tend to be phenomenologically recessive because the phenomenal experience of my action already involves an intentional aspect, and that’s what occupies my attention. Indeed, whatever I am trying to accomplish in the way of basic movements (e.g., moving out of the way, walking to open the door, reaching for a drink)—that is, my intention—informs my body-schematic (motor-control) processes, which are intentional just because they are constrained by what I am trying to do. We’ll return to this idea of motor intentionality shortly.

### 3.1.2 *Intentional aspects in the sense of agency*

Both phenomenological reflection and brain-imaging experiments suggest that the whole story about the sense of agency cannot be told in terms of efferent processes, even reinforced by reafferent signals. The intentional aspects, and the meaning of what I am trying to do (my aim, goal, or intention), and what I actually accomplish in the world, enter into my sense of agency. Several experiments (e.g., Chaminade & Decety 2002; Farrer & Frith 2002) help to distinguish between the purely motor-control contributories to the sense that I am moving my body, and the most immediate and perceptually based intentional aspects of action; i.e., the sense that I am having an effect on my immediate environment. These experiments, however, already introduce a certain theoretical ambiguity, since they fail to make explicit the difference between motor-control aspects and intentional aspects.

For example, in Farrer and Frith (2002), an fMRI experiment designed to find the neural correlates of the sense of agency, subjects are asked to manipulate a joystick to drive a colored circle moving on a computer screen to specific locations on the screen. In some instances the subject causes this movement and in others the experimenter or computer does. The subject is asked to discriminate self-agency and other-agency. Farrer and Frith begin with the distinction between sense of

agency and sense of ownership in motor-control terms of efferent and afferent processes, but in the experimental design they associate sense of agency with the intentional aspect of action; i.e., whether the agent is having some kind of effect with respect to the goal or intentional task—in this case, whether the agent is controlling something on the computer screen. Accordingly, their claim is that the sense of ownership (the sense that it is “my hand moving the joystick”) remains constant while the sense of agency (“I’m manipulating the circle on the computer screen”) changes. When subjects feel that they are not controlling the events on the screen, there is activation in the right inferior parietal cortex and supposedly no sense of agency for the intentional aspect of the action. When the subject does have a sense of agency for what happens on the screen, the anterior insula is activated bilaterally.

Although Farrer and Frith clearly think of the sense of agency as something tied to the intentional aspect of action and not to mere bodily movement or motor control, when it comes to *explaining why* the anterior insula should be involved in generating the sense of agency, they switch back to their original characterization and frame the explanation in terms of motor control and bodily movement.

Why should the parietal lobe have a special role in attributing actions to others while the anterior insula is concerned with attributing actions to the self? The sense of agency (i.e., being aware of causing an action) occurs in the context of a body moving in time and space . . . [and] critically depends upon the experience of such a body. There is evidence that . . . the anterior insula, in interaction with limbic structures, is also involved in the representation of body schema [important for motor control]. (Farrer and Frith 2002, 601–2)

More specifically they point to integrative processes in the anterior insula that involve aspects of both efference and afference.

In particular there will be a correspondence between three kinds of signal: somatosensory signals directly consequent upon our movements, visual and auditory signals that may result indirectly from our movements, and last, the corollary discharge [efferent signal] associated with motor commands that generated the movements. A close correspondence between all these signals helps to give us a sense of agency. (Farrer and Frith 2002, 602)

Likewise, in a separate study, Farrer et al. (2003) have the same goal of discovering the neural correlates of the sense of agency. In this experiment subjects provide a report on their experience; however, all questions about agency are focused on bodily movement rather than intentional aspect. In fact, subjects were not given an intentional task to carry out other than making random movements using a joystick, and the focus of their attention was directed towards a virtual (computer image) hand that either did or did not represent their own hand movements, although at varying degrees of rotation relative to true position of the subject’s hand. That is, they moved their own hand, but saw a virtual hand projected on screen at veridical or non-veridical angles to their own hand; the virtual hand was either under their control, or not. Subjects were asked about their experience of agency for control of the virtual hand movements. The less the subject felt in control, the higher the level of activation in the right inferior parietal cortex, consistent with Farrer and Frith (2002). The more the subject felt in control, the higher the level of activation in the right *posterior* insula. This result is in contrast with the previous study where sense of agency was

associated with activation of the right *anterior* insula. Referencing this difference, Farrer et al. state: “We have no explanation as to why the localization of the activated areas differ in these studies, except that we know that these two regions are densely and reciprocally connected” (2003, 331). One clear explanation, however, is that the shift of focus from the intentional aspect (accomplishing a task in Farrer and Frith) to simple control of bodily movement (in Farrer et al.) changes the aspect of the sense of agency that is being studied. Accordingly, to make sense of these experiments it would help to clearly distinguish between the intentional aspect and the motor aspect of agency, and to say that there are at least these two contributories to the sense of agency (Farrer et al. 2013 do make a clear distinction in this regard).

What we learn from these experiments supports a significant phenomenological point; namely, that even the pre-reflective sense of agency is complex, deriving in part from motor-control processes and in part from a perception of what I am accomplishing by my action (the intentional aspect). Normally these two aspects are tightly integrated. If, for some reason, I lose motor control over my movements, however, I would likely lose my sense of agency (specifically, my sense of control). In a slightly different way, if I start to feel that my actions are not having the intended effect in the world, my sense of agency (specifically, my sense of efficacy) would be diminished.<sup>1</sup> As we’ll see, however, this still does not give us the full picture of the sense of agency.

## 3.2 Intention Formation and Attribution

### 3.2.1 *Intentions*

Over and above the sensory-motor processes that involve motor control, and the perceptual processes that allow us to monitor the intentional aspects of our actions, there are higher-order cognitive components involving intention formation that also contribute to the sense of agency.<sup>2</sup> Pacherie (2007), and others like Bratman (1987) and Searle (1983), distinguish between *distal* intentions (sometimes referred to as future or prior intentions) and *proximate* (or present) intentions. Distal- or D-intentions relate to prior deliberation processes that allow us to formulate our relatively long-term (future) goals. For example, I may decide to purchase a car tomorrow (or next week, or next month, or at some undetermined time in the future), and form the intention to go out and engage in that action when the appropriate time comes. When the time comes, I put my D-intention into action in a very specific way; for example, when I actually go to shop for a car. In this regard, my action reflects a proximate or P-intention. The P-intention specifies the action in terms of the particular requirements of the action situation, including the

<sup>1</sup> See Farrer et al. (2013) for the importance of temporality in this regard; also see Braun et al. (2018) for a review of the experimental literature, and Deans (2019) for how these complexities can affect experimental studies of the sense of agency.

<sup>2</sup> I note that the term “intention” as it is used in the following discussion most often signifies action intention. On my understanding the concept of action intention is not unrelated to the more general concept of intentionality understood in the sense of “aboutness,” as found, for example, in Brentano and the phenomenological tradition. Action intention is one kind of intentionality in this sense; my action intends (is about or is directed towards) something (e.g., a goal).

circumstances tied to particular environments. I decide to go to a particular car dealership; I walk around looking at different cars; I kick tires; I discuss financing; I decide to purchase a specific car. The P-intention is what Searle calls the “intention-in-action,” and I form it as I go, or in an ongoing way, sometimes out of habit or custom, and often in an *ad hoc* way depending on circumstance.

Not all actions involve prior D-intention formation. For example, I may decide right now to get a drink from the kitchen and find myself already moving in that direction. In that case I have not formed a D-intention, although my action is certainly intentional. Rather I may simply form a proximate or P-intention. My intention to get a drink from the kitchen may involve acting on an inclination to get up from my desk and move around and for contingent reasons having to do with the layout of my abode, that means I move in the direction of the kitchen. In doing so I may be monitoring what I am doing in a more or less general way, if, for example, it turns out to be a rather complex action, stepping over children, or around a pile of books. At my university office the kitchen is located down the hall and is locked in the evening. If I want to get a drink I have to walk up the hall, retrieve the key for the kitchen from a common room, and then proceed back down to the kitchen, unlock the door, retrieve the drink, relock the door, return the key, and then return to my office. Although I may be thinking of other things as I do this, I am also monitoring a set of tasks that are not automatic. The ongoing formation of the P-intention keeps the action on track.

In other cases I may be so immersed in my work that I don’t even notice that I’m reaching for the glass of water on the table next to me. Here my intentional action may be closer to habitual action, with no P- or D-intention involved. In such cases, my action is still intentional, and I would still have a pre-reflective sense of agency, connected with what Pacherie (2007) calls a motor or M-intention, and consisting of the above-discussed pre-reflective sense generated in motor-control processes and a rather recessive intentional aspect (which I may only notice if I knock over the glass or spill the drink).

Following Pacherie, then, we can distinguish between three kinds of intention.

- **D(istal) intention:** I deliberate about what I will do at a later point and prospectively form my intention (e.g., considering certain facts, I decide to buy a car next week).
- **P(roximate or Present) intention:** Searle’s concept of intention-in-action (e.g., I reflectively or perceptually monitor/guide my actions in terms of specific means-ends relations as I carry them out to meet a particular goal).
- **M(otor) intention:** involves the control processes that keep an action on track; in relation to a P-intention it’s involved in selecting the movement appropriate for carrying out the intended action (e.g., without conscious perceptual monitoring, I move my arm thus and so in order to pick up the cup and take a drink).

It seems feasible that when there is a D- and/or P-intention involved, such intentions generate a stronger sense of agency. Certainly, if I form a D-intention to buy a new car tomorrow, and tomorrow I go to the car dealership and purchase a car, I will feel more in charge of my life than if, without prior intention I simply find myself lured into a car dealership, and purchasing a car without prior planning. In

the latter case, even if I do not deny that I am the agent of my action, I might feel a bit out of control. So it seems clear that part of the phenomenology of agency may be tied, in some cases, to the prospective formation of a prior intention. It's important here to distinguish between the second-order reflective, prospective type of cognitive process involved in D-intention formation—which may be framed in terms of a narrative, and which may involve making judgments and decisions based on beliefs, desires, or evaluations (see, e.g., Frankfurt 1978; Velleman 2006)—and a first-order (pre-reflective) level of experience where we find the sense of agency as an experience of agency. The sense of agency is not itself a judgment, although I may judge that I am the agent of a certain action based on my sense of agency for it. But what is clear is that intention formation may generate a stronger sense of agency than would exist without the formation of D- or P-intentions.

### 3.2.2 *Retrospective attribution*

The effect of the formation of a prior intention is clearly prospective. But there are post-action processes that can have a retrospective effect on the sense of agency. Graham and Stephens (1994; Stephens & Graham 2000) provide an account of two kinds of introspective self-attribution.

- *Attributions of subjectivity or ownership*: the subject reflectively realizes and is able to report that he is (or was) moving.
- *Attributions of agency*: the subject reflectively realizes and is able to report that he is the cause or author of his movement. For example, he can say “I caused this action.”

Distinguishing between agency and ownership, and between reflective attribution and pre-reflective experience, we can map out differences between sense of agency, sense of ownership, attribution of agency, and attribution of ownership (*Table 3.1*).

In contrast to the idea that the sense of agency is pre-reflective and generated in motor-control processes and intentional aspects, according to Graham and Stephens the sense of agency originates at the higher-order level of attribution. In other words, for them, the sense of agency is equivalent to the attribution of agency. They propose an explanation of the sense of agency in terms of “our proclivity for constructing self-referential narratives” which allow us to explain our behavior retrospectively: “such explanations amount to a sort of theory of the person’s agency or intentional psychology” (1994, 101; Stephens & Graham 2000, 161). If we take thinking itself to be a kind of action on our part, then, on this account, our sense of agency for that thinking action, as for any action, derives from a reflective attitude toward it.

[W]hether I take myself to be the agent of a mental episode [or bodily action] depends upon whether I take the occurrence of this episode to be explicable in terms of my underlying intentional states. (1994, 93)

On this view our sense of agency for a particular action depends on whether we can reflectively explain our action in terms of our beliefs, desires, and D-intentions. Accordingly, if a subject does or thinks something for which she has formed no intention, and her action fails to accord with her beliefs and desires—mental states

Table 3.1 Terminology varies from one theorist to another; sometimes a distinction is made between agency and authorship, and sometimes that distinction is denied. Likewise with respect to the sense of ownership it is sometimes equated with a sense of subjectivity (a sense that I am living through an experience), and sometimes with a feeling of mineness (this is *my* experience), or both, where these two experiences are either distinguished or collapsed.

| Experience                               | Pre-reflective                                                                                                                                                         | Reflective                                                                                                                                            |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Agency<br>(Authorship)                   | <i>Sense of agency</i> —as I act or think, the pre-reflective <i>experience</i> that I am the cause/initiator of my action or thinking, and have some control over it. | <i>Attribution of agency</i> —the reflective (retrospective) attribution or <i>judgment</i> that an action or a thought was caused by me.             |
| Ownership<br>(Subjectivity,<br>Mineness) | <i>Sense of ownership</i> —the pre-reflective <i>experience</i> that I am moving (or keeping still) or thinking—that the movement, action, or thinking is mine.        | <i>Attribution of ownership</i> —the reflective (retrospective) realization or <i>judgment</i> that I am the one who moved or had a specific thought. |

Source: Adapted from S. Gallagher, “Deflationary accounts of the sense of ownership” in: F. de Vignemont and A. Alsmith (eds.), *The Subject’s Matter: Self-Consciousness and the Body*, pp. 145–62, Cambridge, MA, Copyright © 2017 Massachusetts Institute of Technology, published by The MIT Press, Table 7.1, p. 146, doi: 10.7551/mitpress/10462.003.0010.

that would normally explain or rationalize the action—then the action or thought may not appear for her as something she intentionally does or thinks. Whether I count something as my action thus

depends upon whether I take myself to have beliefs and desires of the sort that would rationalize its occurrence in me. If my theory of myself ascribes to me the relevant intentional states, I unproblematically regard this episode as my action. If not, then I must either revise my picture of my intentional states or refuse to acknowledge the episode as my doing. (1994, 102)

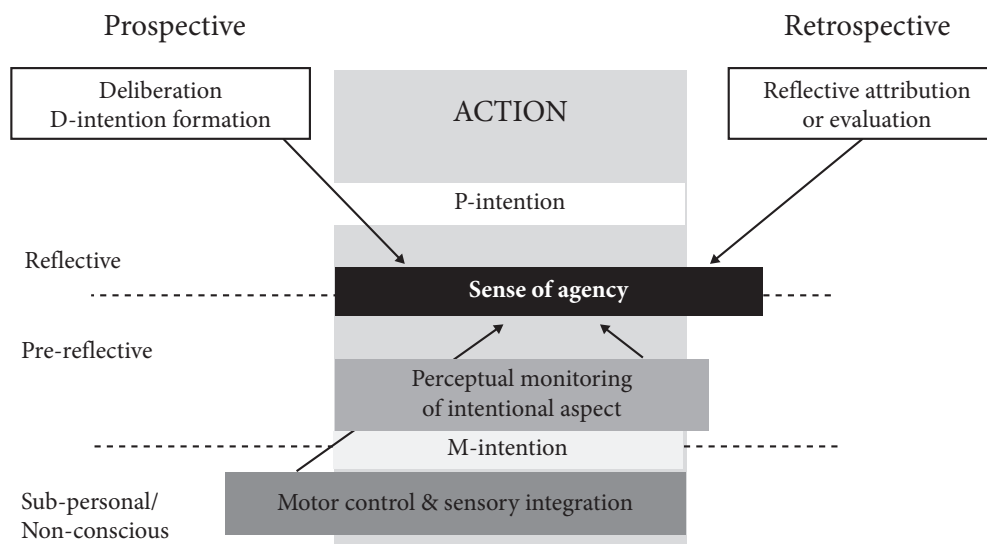
I think this is an overly intellectualized view. One seemingly has a sense of agency because one has a properly ordered set of second-order retrospective interpretations concerning a particular action (see Graham & Stephens 1994, 102; Stephens & Graham 2000, 162ff).

Pacherie indicates that D-intentions are subject to normative pressures for consistency and coherence relative to the agent’s beliefs and other intentions. This would also seem to be the case with Graham and Stephens’ retrospective attributions. The fact that retrospectively I may fail to justify my actions, or that I may think that my actions fail to fit with my theory or my self-narrative, does not necessarily remove my pre-reflective sense of agency for the action. Indeed, on the one hand, one could think that for me to be able to pick out any particular action as one that I would attribute to myself, it would have to be one that I have actually lived through and experienced in an agential fashion, with a pre-reflective sense of agency. In this respect, my attribution of agency may just be a report of that pre-reflective sense of agency that I experienced as I acted. On the other hand, if the action seems to be inconsistent with my self-narrative (e.g., about what sort of person I am and what sort of action

I typically engage in), this modification of failure of attribution may diminish my sense of agency.

My own view is that retrospective attribution (or judgment) of self-agency may have an effect on a continued (temporally extended) pre-reflective sense of agency for my action, either strengthening it or weakening it, but not that my attribution actually constitutes the sense of agency. In this respect Synofzik et al. (2008) rightly distinguish between an experience of agency and a judgment of agency, where the latter is something over and above the pre-reflective sense of agency.

Given these different contributories it seems possible to speak of different (stronger or weaker) degrees of the sense of agency. In regard to some aspects of my life I may generally feel that I am in control of my life because I usually follow through and act on my intentions. I think and deliberate about an action, and form a D-intention to do it. When the time comes I remember my D-intention and I see that it is the appropriate time and situation to begin acting to fulfill that intention. My P-intentions coincide with the successful guidance of the action; my motor control is good and all of the intentional factors line up. There is a good alignment of my D-, P-, and M- intentions. Subsequently, as I reflect on my action, it seems to me to be a good fit with how I think of myself and I can fully attribute responsibility for that action to myself. It seems that in this case I would feel a very strong sense of agency for the action, all contributing aspects—prospective intention formation, contemporary control, and intentional factors, as well as reinforcing retrospective attributions—giving me a coherent experience of that action (see *Figure 3.1*), although this may not always be the case (see Deans 2019). Alternatively, however, I may have a minimal sense of agency because there is no D- or P-intention and no retrospective attribution or evaluation. My sense of agency for the action may just be my thin experience of having motor control over something I am doing. Likewise, I may experience a diminished sense of agency if I am depressed or in cases of anxiety or depersonalization (Gallagher & Trigg 2016; Gentsch & Synofzik 2014).



**Figure 3.1** Complexities in the sense of agency.

### 3.3 Ambiguities

According to some neuropsychologists, the sense of agency and the sense of ownership may be generated by subpersonal comparator mechanisms.<sup>3</sup> According to this comparator hypothesis, motor-control mechanisms associated with M-intentions involve inverse and forward models. An inverse model derives the motor commands required for a particular movement or body position by comparing current body position with desired body position. When the inverse model generates a motor command the efference copy of that command is sent to a forward model, which compares the intended body position with the predicted sensory feedback (from the anticipated body position) that will result from an issued motor command. If there is a mismatch (or prediction error) in the system, the forward model allows for quick, online correction as the action is in process, without waiting for sensory feedback from the movement. Pacherie suggests that D- and P-intention formation can also be analogized to these kinds of forward and inverse motor-control mechanisms (2007, 4).

That our deliberation about future actions involves thinking about the means and ends of our actions seems uncontroversial, although it is not necessarily the case that when I form a D-intention I consciously employ an inverse model to determine the means or the strategy I will follow to achieve it. Pacherie's proposal does raise an important question, however. If we regard thinking, such as the deliberative process that may be involved in intention formation, as itself a kind of action, then do we also have a sense of agency for the thinking or deliberation involved in the formation of D-intentions? It seems right to suggest that if I engage in a reflectively conscious process of deliberating about my future actions and make some decisions on this basis, I would have a sense of agency for this deliberation. You could interrupt me during this process and ask what I am doing, and I could say: "I'm sitting here deliberating about buying a car." The sense of agency that I feel for my ongoing deliberation process may be based on my sense of control over it; my response to your question is a retrospective attribution that may confirm this sense of agency. It's also possible that my sense of agency for my deliberation derives in part from a previous deliberation process; for example, I may have formed the D-intention yesterday to do my deliberations (i.e., to form my D-intentions) about car buying today. It is clearly the case, however, that not all forming of D-intentions require a prior intention to do so, otherwise we would have an infinite regress. We would have to deliberate about deliberating about deliberating, etc. Furthermore, it is possible to have P-intentions for the action of forming D-intentions, where P-intentions in this case may be a form of metacognition where we consciously control our cognitive strategies as we form our D-intentions (e.g., I first better consider reasons for and against, and I won't worry about what my wife might say about purchasing a car until

<sup>3</sup> When I use terminology associated with comparators, forward models, efference copy, emulator, etc. in the context of motor-control processes, it is for purposes of discussion. I don't mean to endorse such models. These are standard concepts used by many theorists, but there are alternative ways of explaining such processes; e.g., dynamical systems theory or notions of synergy, more consistent with my enactivist inclinations. See, e.g., Chemero & Turvey (2007); Kelso (1995); Turvey (2007). Thanks to one of OUP's reviewers for a caution in this regard.

later). Certainly, however, it is not always the case that we engage in this kind of metacognition as we formulate our D-intentions. It seems, then, that we can still have a minimal first-order sense of agency for our deliberations without prior deliberation or occurrent metacognitive monitoring.

On the one hand, my sense of agency for actually carrying out a particular action (X) is different from my sense of agency for the process of forming the D-intention to do X. They are obviously not equivalent since there are two different actions involved, X and the act of deliberation about X. On the other hand, it seems likely that the sense of agency for my deliberation may contribute to my reflective sense (including my retrospective attribution) that I am the agent of my own actions. This relates to what Pacherie refers to as the long-term sense of agency:

a sense of oneself as an agent apart from any particular action, i.e. a sense of one's capacity for action over time, and a form of self-narrative where one's past actions and projected future actions are given a general coherence and unified through a set of overarching goals, motivations, projects and general lines of conduct. (2007, 6)

As such this long-term sense of agency may enter into and reinforce the occurrent sense of agency for any particular action. Furthermore, if I lacked a sense of agency for my deliberation process, that process might feel more like an intuition or unbidden thought, or indeed, if I were schizophrenic, it might feel like an inserted thought. In any case, it might feel less than integrated with what Graham and Stephens call the "theory or story of [my] own underlying intentional states," something that itself contributes to the sense of agency for the action. So it seems that the sense of agency for the deliberation process itself may contribute to the sense of agency for the action X in two indirect ways. First, by contributing to my long-term sense of agency, and second, by contributing to the effect of any retrospective attribution I may engage in. Still, as I indicated, there need not be (and, under threat of infinite regress, there can not be) a deliberation process for every action that I engage in.

The case is similar for P-intentions. If action monitoring, at the level of P-intentions, is itself a kind of action (if, for example, it involves making judgments about or actively directing my attention to certain environmental factors), there may be a sense of agency for that action monitoring. The processes that make up a P-intention, however, are much closer to the intended action itself and may not feel like an additional or separate action. Yet there could be a very explicit kind of P-intention in the form of a conscious monitoring of what I am doing. For example, I may be putting together a piece of furniture by following a set of instructions. In that case I could have a sense of agency for following the instructions and closely monitoring my actions in terms of means-ends. Certainly doing it that way would feel very different from doing it without following the set of instructions. But the sense of agency for following the instructions would really go hand in glove with the sense of agency for the action of assembling the furniture. How we distinguish such things would really depend on how fine-grained we get in defining the action.

In the process of assembling the furniture, I may start by reading instruction #1; I then turn to the pieces of wood in front of me and join two of them together. I can distinguish the act of reading from the act of joining, and I can delineate a sense of

agency for each of them. In that case, however, one can ask whether my sense of agency for the act of reading doesn't contribute to my sense of agency for the act of joining. I might, however, think of the reading and the joining as one larger action of assembling the furniture, and my sense of agency might be defined broadly in a way that pertains to all aspects of that assembling. It might also be the case that when I put together a second piece of furniture I remember rather than read the instructions, and when I put together the *n*th piece of furniture I don't follow instructions at all, in which case the sense of agency is more concentrated in the joining. In most practiced actions a P-intention is really unnecessary because motor-control processes and a minimal perceptual monitoring of the intentional aspect (especially based on specific environmental or artifactual features) can do the job; i.e., can keep my action on track. I might simply make up my mind to do this habitual task, and I go and immediately start to do the task without further monitoring in terms of means-ends. All of this suggests that how we experience agency is relative to the way we individuate specific actions, and how practiced those actions are.

This means that there is some serious ambiguity, not simply in the way we define the sense of agency, but in the sense—in the experience—of agency itself. This phenomenological ambiguity—the very ambiguity of our experience of agency—should be included in any considerations about the sense of agency. Clear-cut and unambiguous definitions may create a neat conceptual map; but the landscape itself may not be so neat. It is not always the case that P-intentions serve to implement action plans inherited from D-intentions, since there are not always D-intentions. It is not always the case that “the final stage in action specification involves the transformation of the perceptual-actional contents of P-intentions into sensorimotor representations (M-intentions) through a precise specification of the spatial and temporal characteristics of the constituent elements of the selected motor program” (Pacherie 2007, 3), since there are not always P-intentions associated with actions. Pacherie also suggests that a sense of action initiation and a sense of control are “crucial” components in the sense of agency (2007, 17–18) and that in both components the P-intention plays a large role. But the fact that some actions for which we have a sense of agency take place without P-intentions puts this idea in question.

The *sense of action initiation*, Pacherie suggests, is based on the efference binding of the awareness of P-intention and awareness of movement onset in the very small time frame of 80–200 msec prior to actual movement onset. This corresponds to the time of the lateralized readiness potential, a neuronal signal that seemingly corresponds to selection of a specific motor program (Libet 1985; Haggard 2003) or to motor execution (Schurger, Sitt, & Dehaene 2012; but see Schlegel et al. 2013). Pacherie associates the P-intention with what Patrick Haggard distinguishes as the urge to move and a reference forward to the goal of the action. These aspects of action experience, however, generated by pre-motor processes, more likely correlate with the M-intention (see Desmurget et al. 2009 for relevant data). In this regard it is important to distinguish P-intention from more basic motoric processes that can occur without a formed P-intention, as in practiced action. Even in practiced, habitual, or skilled action I can have a sense of action initiation without involving a conscious monitoring or P-intention. Since intentional control of action can contribute to the sense of agency whether we have a conscious monitoring in

terms of specific means to goals or not (Aarts, Custers, & Wegner 2005), engaging a well-formed P-intention does not seem crucial for the sense of agency.

Pacherie (2007) further suggests that the *sense of control*, as a component of the sense of agency, has three dimensions corresponding to the rational control aspects of D-intentions, the situational control aspects of P-intentions, and the purely motor-control aspects of M-intentions. Again, however, a number of qualifications can be made given that D- and P-intentions are not involved in all cases of action, and the sense of control associated with the M-intention may be very recessive as long as the action is going well; e.g., as long as I don't stumble over or knock into something. In some cases, the engagement of a P-intention—i.e., the monitoring of specific means to goal, or the conscious navigation through a complex environment, and a more explicit conscious sense of control (or loss of control) are absent until motivated at that point when something starts to go wrong at the motor-control level. The same might be said about what Terry Horgan (2017, 157) calls the feeling of “optionality”—the feeling that you could veto the action or that you could have done otherwise. This may be an intrinsic dimension of the phenomenology of agentic control, but not one that necessarily stands out in our experience.

More generally, what seem to be legitimate conceptual distinctions on the theoretical level—“awareness of a goal, awareness of an intention to act, awareness of initiation of action, awareness of movements, sense of activity, sense of mental effort, sense of physical effort, sense of control, experience of authorship, experience of intentionality, experience of purposiveness, experience of freedom, and experience of mental causation”—all of which Pacherie (2007, 6) suggests are distinctions that can be found in the phenomenology of the sense of agency—may not in fact show up as such in the actual first-order phenomenology. Indeed, in any particular case, some of these aspects may be the product of theoretical reflection on the first-order phenomenology rather than actual aspects of the experience. As I engage in action, for example, I may not experience a difference between my sense of effort and my sense of control, although I can certainly make that distinction in my reflective consideration of my action. That distinction may show up clearly at the level of my retrospective attribution, but may be entirely absent in my current, situated pre-reflective sense of agency. As previously indicated, my awareness of what I am doing and that I am doing it is usually struck at the most pragmatic level of description (“I’m getting a drink”) rather than at a level that distinguishes between the action and my agency, or within the action between the goal and the means, or within agency between intentional causation, initiation, effort, purposiveness, control, and so forth. The sense of agency is complex, but it may not be *that* complex.

At another extreme, we should resist an overly simplified phenomenology of agency. Phenomenologically, for example, there is no such thing as a “naked intention”—that is, the idea that, in the case of our own action, we can be aware of the action without being aware of *who* the agent is (Jeannerod & Pacherie 2004)—or what Pacherie (2007, 16) calls “agent-neutral” action experience. This concept confuses neurological and phenomenological levels of analysis. It is possible to identify neurons that are activated both in the case of one’s own action and in the case of another person’s action. Mirror neurons (MNs) are an example. In this regard, it’s often claimed that MNs, by themselves, are neutral with respect to

who the agent is—they do not distinguish whether oneself or the other is acting (Vignemont 2004; Gallese 2005; Hurley 2005; Jeannerod & Pacherie 2004).<sup>4</sup> This idea motivated Georgieff and Jeannerod (1998) to posit a mechanism called the “Who system” to account for the determination of agency. On this model, if some set of neurons (e.g., mirror neurons) fire in both cases of self- and other-agency (that is, if there is an overlap of neuronal activation that registers intentional action), there must be, in addition, other neurons activated in the non-overlapping areas that signal that the action is mine rather than yours, or vice versa. This mechanism, according to Georgieff and Jeannerod, answers the question: “Whose action is it?” Although there is no claim that these processes are not simultaneous, one might conceive of such processes as involving two logically distinct steps: activation for intentional action (without agent specification—thereby neutral or naked) plus activation for agent identification.

Jeannerod and Pacherie assume that an articulation at the level of neural activations, specifically between (1) those activations responsible for registering the “naked” intention of an action (e.g., this is an action of opening the door), and (2) those activations responsible for registering the agent for the action, means that there is an articulation *in experience* between awareness of intention and the experience of agency. They suggest: “We can be aware of an intention, without by the same token being aware of whose intention it is . . . [S]omething more than the sole awareness of a naked intention is needed to determine its author” (Jeannerod & Pacherie 2004, 140). Even if it were possible for the brain to process information about intentions distinct from assigning an agent to the intentions, is it legitimate to say that our experience is similarly articulated? Jeannerod and Pacherie suggest that it is.

When the naked intention one is aware of yields an overt action, the extra information needed to establish authorship may be found in the outside world. The question “Is this intention mine?” would then be answered by answering the question: “Is this my body performing the corresponding action?”. (140)<sup>5</sup>

Phenomenologically, however, intentions in every case come already clothed in agency.<sup>6</sup> The “who” question, which is rightly posed at the neurological level, doesn’t

<sup>4</sup> There are good reasons to doubt that this is the case. See *Section 4.3.6*.

<sup>5</sup> This extremely articulated process, diagrammed in the form of answering separate questions, is set out in an explanation that resembles Brentano’s theory of intentionality and self-awareness. On this view, perception of a tree involves primary and secondary objects. The primary perceptual object is the tree; the secondary object is myself as perceiver. “We claim that it is like this with the perception of intention: when Mary watches John open the door, she is primarily aware of an intention to open the door, rather than being primarily aware that John intends to open the door. Similarly, when Mary herself intends to open the door, she is primarily aware of an intention to open the door, rather than being primarily aware that she herself intends to open the door. Let us call this awareness of an unattributed or ‘naked’ intention” (Jeannerod & Pacherie 2004, 116). For a critique of Brentano’s theory of intentionality, see Gallagher and Zahavi (2012). Even if Brentano were right about self-awareness, the idea that we perceive naked intentions in another’s action, and secondarily attribute agency, seems to be based on a supposed isomorphism between subpersonal and personal-phenomenological levels.

<sup>6</sup> Even in schizophrenic delusions of control, the subject does not experience naked intentions; he attributes agency to someone else, even as he experiences the movement as his own, in the sense that it is his own body moving. Indeed, in such a case, the question “Is this intention mine?” is not asked as a separate question, or answered by answering the question: “Is this my body performing the corresponding action?”

come up at the level of experience, because the neural system has already decided the issue—one way or the other. That is, even if I'm wrong about who the agent of an action is (something that may happen even in regard to one's own action in schizophrenic symptoms of delusions of control), I still experience or perceive the intended action as already determined with respect to who the agent is.

A similar problem arises with claims that the phenomenology of the sense of agency and the sense of ownership are inflationary (see, e.g., Bermúdez 2011; Grünbaum 2015); i.e., that a positive sense of agency or sense of ownership does not arise until something goes wrong with action, or that it is generated as the result of reflection. Thor Grünbaum (2015), for example, suggests that there is no separate and distinct pre-reflective sense of agency that acts as the basis for a judgment about agency. He specifically challenges the idea that comparator processes generate a distinct experience of agency, and he takes the claim that the sense of agency is generated by such mechanisms to mean that the sense of agency is intention-free. That is, on the comparator model, the sense of agency is generated even if the agent has not formulated a prior, personal-level D-intention to act in a certain way. As we've seen, however, a D-intention is not the only kind of intention, and is not always involved in action. Reaching for my cup of tea as I work on my computer does not require that I have a prior intention to do so, or that I consciously deliberate and form a plan to do so. Yet it still counts as an intentional action and may involve a present intention-in-action, and motor intentions (Pacherie 2006; 2007). Moreover, at least some comparator models include the idea that there is some functional element in the system that counts as an intention, and that this intention is compared to efference copy or sensory input from the movement to facilitate motor control (e.g., Frith 1992; Wolpert & Flanagan 2001). In this respect it's not clear that even on the comparator model, the sense of agency can be characterized as intention-free.

Grünbaum may be right about comparator mechanisms not generating the sense of agency; indeed, there are other models, and good reasons to question whether comparator models of motor control offer the best explanation (see, e.g., Friston 2011; Synofzik, Vosgerau & Newen 2008). That issue aside, however, the pre-reflective sense of agency may be constituted by a number of contributories, including not only motor-control aspects, but also the pre-reflective intentional aspect discussed above; that is, one's perceptual monitoring of what one's actions are accomplishing in the world.

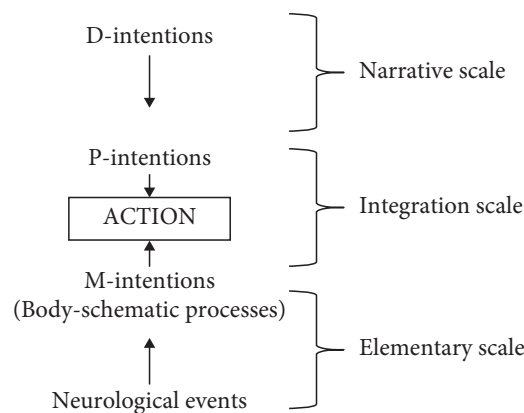
If actions were intention-free, or agent-anonymous (i.e., characterized by naked intentions), or if we did not have an intrinsic sense of agency and sense of ownership for our actions,<sup>7</sup> we would be constantly puzzled by our actions, or asking: did I just

<sup>7</sup> Bermúdez (2011), for example, denies that there is a positive first-order phenomenology of ownership or non-observational (or pre-reflective) feeling of "mineness." He takes the sense of "mineness" to be inferential or a result of introspective attribution. It's not clear, however, that Bermúdez would deny the idea that we have a proprioceptive and kinesthetic awareness of our bodily (and limb) posture and movement. By definition, proprioceptive/kinesthetic awareness is experience of my own body—an intrinsic experience that it is *my own* body that is positioned such and such, or moving so and so. That's the *proprio* in proprioception. A recessive/attenuated (uninflated) sense of ownership or feeling of mineness, however, does not include anything more than that; rather it is an intrinsic structure of experience itself. See the notion of a "perspectival" sense of ownership (Albahari 2016); i.e., a minimal sense of ownership

do that, or was that you? The fact that such puzzles are rare, and sometimes considered pathological when they do occur (as in schizophrenic delusions of control), supports the phenomenological view that these aspects of pre-reflective self-awareness typically are intrinsic aspects of experience (see Gallagher, 2017b; Gallagher & Trigg 2016; Gallagher & Zahavi 2012). In this respect, although Peter Langland-Hassan (2008) raises similar worries as Grünbaum about the positive phenomenology of the sense of agency, he concludes that the phenomenology of agency is “one that is embedded in all first order sensory and proprioceptive phenomenology as diachronic, action-sensitive patterns of information; it does not stand apart from them as an inscrutable emotion” (392).

In *Chapter 2* I focused on the intrinsic temporal structure of action at the elementary and integrative timescales and I explored the micro-structures that characterize subpersonal processes and the pre-reflective level of experience. I argued for a dynamical enactive account of these phenomena. At this point it should be clear that the threefold distinction among elementary, integration, and narrative timescales usefully maps onto the three kinds of intention or intention formation discussed in this chapter (see *Figure 3.2*). The elementary scale correlates to the motor-control processes (including forward and inverse models) that define M-intentions. The integration scale corresponds roughly to P-intentions, which involve quick decisions and ongoing monitoring of actions in the detailed parameters of the current environment. The narrative scale involving longer-term memory, deliberation, and planning corresponds to D-intentions.

Pacherie is right to note that a conceptual analysis cannot “preempt the question whether these various aspects are dissociable or not, for instance whether we can be aware of what we are doing independently of an awareness of how we’re doing it or whether we can be aware of what we are doing without at the same time experiencing this action as ours” (2007, 7). What can decide the issue, however, is agreement on where to draw the lines between *phenomenological* analysis (i.e., of what we actually



**Figure 3.2** Intentions and timescales.

connected to the fact that such experiences have an intrinsic structure directly tied to a first-person perspective. See Gallagher (2017c); Gallagher and Zahavi (2012).

experience), *neuroscientific* analysis (which, as we've seen in the non-linear relations between elementary and integrative time scales [Chapter 2], may find a much finer grain of articulations at the neuronal level than what shows up in phenomenology), and *conceptual* analysis (which may introduce distinctions that are in neither the phenomenology nor the neurology, but may have a productive role to play in constructing cognitive models or, in regard to the individual, explaining psychological motivations, or forming personal narratives, etc.).

I suggest that when we integrate these analyses in the right way, we find that the sense of agency and processes of intention formation are complex and ambiguous. Some aspects of action can involve cognitive and conscious reflection and deliberation; many other aspects of action are non-conscious. Likewise, as suggested in previous sections, the sense of agency has multiple contributories, some of which are reflectively (prospectively or retrospectively) conscious, some of which are pre-reflectively conscious, and some of which are non-conscious. Consistent with the phenomenology of embodied experience, in everyday engaged action reafferent or sensory-feedback signals are attenuated, implying a recessive consciousness of the body in action (see, e.g., Gallagher 2005a; Tsakiris & Haggard 2005). We do not attend to the details of our bodily movements in most actions. We do not stare at our own hands as we decide to use them; we do not need to look at our feet as we walk, we do not attend to our arm movements as we engage the joystick.<sup>8</sup> Although efferent, motor-control, and body-schematic processes are non-conscious, just such processes nonetheless contribute to a conscious sense of agency by generating a pre-reflective embodied awareness of our actions.

For most of our everyday actions, then, the sense of agency runs along with and is experientially indistinguishable from a basic sense of ownership, consistent with the integration of efferent and reafferent signals in the insula and other brain areas. The sense of agency involves, in part, a basic feeling of embodiment without which actions would feel very different. In addition, we also experience, pre-reflectively, a form of *intentional feedback*, which is not afferent feedback about our bodily movements, but a perceptual sense that my action is having an effect in the world. This effect is not something that we need to reflectively dwell on, or even retain in memory, especially if there is a consistency between the action intention and what the action actually accomplishes.

The sense of agency for most of our actions may amount to nothing more than these embodied and pre-reflective processes. In some cases, however, we may be reflectively conscious of and concerned about what we are doing and why we are doing what we're doing. We may feel good or bad about our actions. For such actions the sense of agency may become more reflective, tied more directly to a D-intention, involving attention directed toward the project or task that we are engaged in, or toward the means and/or end that we aim for.

<sup>8</sup> Deafferentation, as in the case of IW, counts as an exception to this kind of lack of conscious monitoring. See Cole (1995); Gallagher (2005a); Gallagher and Cole (1995). Also, in some cases of expert performance there may be different kinds of self-awareness involved (Gallagher & Ilundáin-Agurruza, in press).

Conceptually, then, we can identify at least five different contributories to the sense of agency that may be connected with a particular action.

- Formation of D-intentions, often involving a prospective reflective deliberation or planning that precedes some actions
- Formation of P-intentions; that is, the conscious monitoring of action in terms of specific means-ends relations in specific situations
- Basic efferent motor-control processes that generate a first-order experience linked to bodily movement in and towards an environment
- Pre-reflective perceptual monitoring of the effect of my action in the world
- The retrospective attribution of agency that follows action

We can add to this the long-term sense of one's capacity for action over time, related to self-narrative, where past and projected actions provide a general coherence "unified through a set of overarching goals, motivations, projects and general lines of conduct" (Pacherie 2007, 6).

Although conceptually we may distinguish between different cognitive levels (first-order, second-, or higher-order), or aspects (reflective, pre-reflective, non-conscious), and neuroscientifically we may be able to identify different brain processes correlated with these different contributories, in action and in our everyday phenomenology we tend to experience agency in a more holistic, qualitative, and ambiguous way which may be open to a description in terms of degree. The conceptual articulation of the different aspects of the sense of agency, however, suggests that the loss or disruption of the sense of agency in different pathologies may be varied. In schizophrenic delusions of control, for example, the motor-control aspects may be disrupted. In other cases the attribution of self-agency may be disrupted by problems with retrospective higher-order cognition or the prospective formation of D-intentions. A good example of this is the case of narcotic addiction, as discussed by Frankfurt (1988). If a drug addict invests himself in reflectively resisting drugs he may feel that something other than himself is compelling him to drug use. If he withdraws from taking the drug, when he starts using again he may not experience himself as the agent.

It is in virtue of this identification and withdrawal, accomplished through the formation of second-order volition, that the unwilling addict may meaningfully make the analytically puzzling statements that the force moving him to take the drug is a force other than his own, and that it is not of his own free will but rather against his will that this force moves him to take it. (Frankfurt 1988, 18; see Grünbaum 2010, for discussion)

In effect, the sense of agency may be present or absent, diminished or increased depending on processes, or disruptions of processes, at different levels and across different timescales. The degree of presence/absence, and the experiential feel of agency in any case (pathological or non-pathological) may depend on a range of factors, from very specific neurological processes to the type of action in which we are engaged. As I'll suggest in the next section, however, the kinds of ambiguities that we can discover in considering the details of actions and the sense of agency may be either intensified or smoothed over by the circumstances of larger social arrangements.

### 3.4 Pushing this Analysis into the World

So far, although we have acknowledged a role for the body and the environment in action, including many of the pre-reflective aspects being generated in motor control and the intentional aspects of what we are accomplishing in the world, one might argue that almost all of the processes described in previous sections remain “in the head,” insofar as they are considered either mental processes, such as deliberation, intention formation, judgment, evaluation, perceptual monitoring, or brain processes, such as efferent commands, integration of afferent signals, premotor processes, and motor control. On such analyses it seems as if all of the action, all of the important processes concerning intention and action, take place in the narrow confines of the mind or brain, even though we know that action takes place in the world, and most often in social interactions. I want to challenge this kind of narrow internalism or methodological individualism; i.e., the idea that actions, or indeed social processes, are best explained by identifying some mechanism or set of mechanisms within the individual agent. To be clear, the argument depends on a conception of *dynamical constitution* in which many elements come into relations of reciprocal causal couplings (Gallagher 2018a; see *Section 1.1*). This does not eliminate a role for brain processes, but it takes them to work only in dynamical relations with bodily processes and factors best described as elements of the physical and social environment.

How do other people and social forces enter into intention formation, meaning generation, and the sense of agency? This turns out to be a very large question that pushes the analysis towards a host of other issues. To some extent, the remainder of this book is concerned with those issues. For now, and as an initial indication, we can point to some basic ideas.

On the very basic pre-reflective level, the presence of others has an effect on what my possibilities for action are, and the way that I perceive the world in terms of action affordances. Jean-Paul Sartre points in this direction. In his example he is sitting alone in a park. Someone else then enters the scene.

Suddenly an object has appeared which has stolen the world from me. Everything [remains] in place; everything still exists for me; but everything is traversed by an invisible flight and fixed in the direction of a new object. The appearance of the Other in the world corresponds therefore to a fixed sliding of the whole universe, to a decentralization of the world which undermines the centralization which I am simultaneously effecting. (1956, 255)

This somewhat dramatic philosophical description is supported by some interesting science. Consider the Social Simon Effect. The Simon Effect is found in a traditional stimulus-response task. Participants are asked to respond to different color stimuli, pressing a button to their left with their left hand when blue appears and a button to their right with their right hand for red. They are asked to ignore the location of the color (which may be displayed either in their right or left visual field). An incongruence (mismatch) of right *versus* left between the color location and hand used to respond results in increased reaction times (Simon 1969). That is, my response is slower when, for example, red appears on the left rather than on the right and I have to indicate this by pushing the right button with my right hand. When a subject is

asked to respond to just one color with one hand, as you might expect there is no conflict and no effect on reaction time. The surprising thing is that when the subject has exactly the same task (pushing one button for one color) but is seated next to another person who is responding to a different color—each person responding to one color—each acting like one of the hands in the original experiment—reaction times increase for the incongruent trials (Sebanz, Knoblich, & Prinz 2003; Sebanz et al. 2006; Takahama, Kumada, & Saiki 2005). The presence of the other agent seemingly has an effect on my action even in this very simple task.

The Social Simon Effect is controversially the subject of ongoing debates (see Dolk et al. 2014 for a good review). On the one hand, some evidence suggests that the effect is really the result of the sensory-motor contingencies involved with the spatial arrangements of the experiment, and is not really a social effect (Dolk et al. 2011; 2013). On the other hand, there is also evidence that the effects are even more complex and that the specific nature of the interpersonal relationship can affect the results. A cooperative relationship with the co-actor can strengthen the effect, whereas competitive co-actors can abolish or significantly decrease the effect (Hommel et al. 2009; Iani et al. 2011). The effect is also strengthened if the co-actors are romantic (indeed, passionate love) partners, compared with non-romantic partners (Quintard et al. 2018). Even cultural factors may modulate the effect since it is increased among members of a collectivistic religion (Colzato, de Bruijn, et al. 2012 & Colzato, Zech, et al. 2012). It is also increased when there is perceived similarity (*versus* dissimilarity) between co-agents (Müller et al. 2011) and when the subject takes the other to be acting intentionally or as an intentional agent (*versus* non-intentional robot) (Stenzel et al. 2014).

Slowing of reaction times are not necessarily conscious for the agent, and they may have only a minimal effect on one's pre-reflective sense of agency. In some circumstances, however, they may become much more explicitly self-conscious. Consider instances where you are quite capable of and perhaps even proficient at doing action A; for example, successfully free throwing a basketball through the hoop. Your performance may be detrimentally affected, however, simply by the fact of having an audience of very tall basketball superstars. You might in fact feel a degree of inadequacy in such a circumstance, simply because certain people are present.

More generally, the prospective and retrospective dimensions of intention formation and action interpretation, which can have an effect on one's sense of agency, are often shaped by others, and by the situations in which one encounters others. My deliberations to buy a certain kind of car (or any other commodity) may be influenced by what my friends consider to be an appropriate choice. In contrast to internalist views—for example, where simply having a belief about action A encompasses the motivation to A (e.g., Nagel 1970)—and in contrast to many analyses of agency in philosophy of mind and action theory, deliberations, intentions, and motivations to act are not simply mental states (propositional attitudes), or causal brain states—they are often co-constituted with others, and extend beyond any one individual. My decision to buy a new car might be a decision that I've made in conversation with my family. Phenomena such as peer pressure, social referencing, which may be implicit or explicit, or our normatively shaped habitual behavior when in the presence of others, as well as the formation of shared

or collective intentions—these phenomena may detract from or increase one’s feeling of agency.

In this regard, there are extreme cases, like compulsive or addictive situations. In addictive behavior, for example, there is a loss of the sense of agency for one’s actions—but this is not just the result of chemically induced dependency. Compulsive drug-related behaviors correlate neither with the degree of pleasure reported by users nor with reductions in withdrawal symptoms as measured in placebo studies and the subjective reports of users. Robinson and Berridge (1993; 2000) propose the “incentive-sensitization” model: pathological addiction correlates highly with the *salience* of *socially situated* drug-related behaviors and stimuli. For example, specific situations (including the agent’s perception of his social world) are altered, and become increasingly hedonically significant to the agent.

To be clear, to challenge internalism and methodological individualism is not to deny that there are some internal (brain) processes involved in such phenomena, or that the individual agent is not playing some role in the action. Rather, it’s simply to deny that the only thing of importance is the brain’s or the individual’s processes, and to affirm that in some cases a conception of *dynamical constitution* is at stake, such that the reciprocal-causal couplings of body and environmental features contribute in a constitutive (and not just an enabling) way to such phenomena (Gallagher 2017a, 2018a). The fact that the brain regions mediating incentive-sensitization are the same areas that process action specification, motor control, and social cognition—thus, regions of the brain thought to be involved in action-oriented intention formation, social navigation, and action—suggests that the explanatory unit is brain-body-environment rather than just brain (see Miller, Kiverstein & Rietveld 2020). This reinforces the idea that situational saliency—including perceptual saliency of the social situation—contributes to intention formation and the sense of agency—sometimes enhancing, but also (as in extreme addictive behavior) sometimes subverting the sense of agency. Intentions can be dynamically shaped by culturally defined situations, by how others are behaving, and by what may be deemed acceptable behavior within specific subcultures.

The importance of both the meaning context of actions (related to environmental saliences) and their social dimension can be seen even in some of the simplest actions that we perform. This takes us back to some of the studies discussed in *Chapter 1*, showing the importance of meaningful action and socially contextualized action in the context of rehabilitation (e.g., Brouchon et al. 1986; Leontiev & Zaporozhets 1960; Marcel 1992). In many of these cases rehabilitation improved when nominally the same movements were performed in a social situation, in which the movements constituted actions with personal and culturally derived signification. The *socially contextualized* situation is clearly marked by social and personal dimensions that involve norms and cultural categorizations of activities in terms of self-other relations (Gallagher & Marcel 1999).

Another example is both clear and dramatic. It involves congenitally deaf users of American Sign Language (ASL) who suffer from hemispatial neglect following stroke. In cases of left-hemispatial neglect subjects entirely ignore their left perceptual and/or motor field, or the left side of their body. Such patients fail to refer to things in their left hemispace and have a tendency to fail to perceive or attend to what is to

their left. They obviously experience difficulty with spatial tasks that are in their left hemispace or that refer to it; so, for example, they may be unable to map out a room or properly describe the layout of objects in place (Jeannerod 1987). It is important to note that some of the syntactical and discursive aspects of ASL make use of peripersonal space. For example, when entities are introduced in discourse they may be assigned a spatial location around the signer to which the signer returns for anaphoric (especially pronominal) reference. The same ASL users with acquired hemispatial neglect who ignore their left side and have profound difficulties with tasks in their left perceptual field have little to no problem in discursive or syntactical use of left hemispace when involved in socially contextualized communicative acts of signing (Poizner, Klima, & Bellugi 1987). Peripersonal space in this case does register in perceptual or instrumental terms (for recognition or for reaching or grasping, for example), but only as relevant to the agent's intention to represent the reference of discourse in a culturally determined sign system in the conversational situation where it affords communicative possibilities. Such communicative practices are clearly related to social cognition, but the social aspect may be even clearer in the fact that hemispatial neglect patients unable to register objects in their left hemispace are able to identify those very same objects when they are asked to say what another person sees in that space (Becchio et al. 2012).

Importantly, it's not simply that an action intention is shaped by a specific kind of behavioral or social environment. It also goes the other way: action intentions can define or redefine the meaning of the agent's circumstances and how they perceive the environment. It's also the case that the availability and execution of an action is influenced by the description or the intention under which the action is generated. On the one hand, personal goals and motives, as well as the social pragmatics of the situation, can clearly provide the energizing dynamics that smoothly transform intentions into actions; on the other hand, social and personal dynamics can easily be disrupted by any number of factors.

Let me conclude with one further example to indicate how such factors can scale up to affect the long-term sense of agency. Several years ago my daughter Laura worked in the Peace Corps in South Africa, focusing her efforts on HIV education. She recounts that her attempts to motivate residents in a small village outside Pretoria to help themselves by engaging in certain activities in organizing their local communities were met by a sardonic attitude and even polite laughter. They explained to her that such ideas were impossible for them; they were unable to help themselves simply because, as everyone already knew, they were lazy. That's just the way they were, they explained, and they knew this because all their life, under the apartheid regime, they had been told so by various educational and governmental institutions. In effect, because of certain long-standing racist social arrangements, arrangements with prolonged effects even after the dismantlement of apartheid, they had no long-term sense of agency. Social-political-institutional arrangements thus robbed them of possibilities for action, reduced their social and cultural affordances, as well as their sense of agency.

If, to understand action, agency, and intention formation we need to consider the most relevant pragmatic (or highest affordance) level from the perspective of the agent, there is strong evidence that the most relevant pragmatic level is not the level

of internal mental or brain states. We shouldn't be looking exclusively inside the head. Rather, embodied action happens in a world that is physical and social and cultural and political; a world that reflects not only perceptual and affective valiances, but also the effects of forces and affordances that are normative and social. Notions of agency and intention, as well as notions of autonomy and responsibility, as I'll try to make clear later, are best conceived in terms that include social-cultural effects. Intentions often get co-constituted in interactions with others—indeed, some kinds of intentions are simply not reducible to processes that are contained exclusively within one individual. In such cases, the sense of self-agency will be a matter of degree; it can be enhanced or reduced by physical, social, economic, and cultural factors that work their way into and through our own practices, but also, in pathological cases, for example, by loss of motor control or disruptions in pre-reflective action-consciousness.

To return to the question I started with: What is action if it is something more than bodily movement? I began by suggesting that action, in the sense of autonomous behavior which scales up to more than a set of causal processes, is characterized by intention, a sense of agency, and meaning, all of which go beyond simple bodily movement. At the same time, intention (specifically M-intention) and the sense of agency (specifically, the pre-reflective sense of agency that is generated in motor-control processes) are not detached from bodily movement. Action carries movement into the realm of meaning and reason; it imbues movement with meaning and intentionality. In this section I've pushed further to provide evidence that intention formation, the sense of agency, and the meaning of action are significantly affected by factors that involve social contextualization; that is, by dynamical arrangements that are intersubjective, normative, cultural, and/or political.